

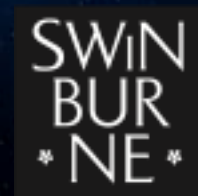
Discovering the Origins of Gravitational-Wave Sources

COMPAS Meeting

Ashwathi Nair

Primary Supervisor : Dr. Simon Stevenson

Secondary Supervisor : Prof. Jarrod Hurley



Centre for
Astrophysics and
Supercomputing

Discovering the Origins of Gravitational-Wave Sources

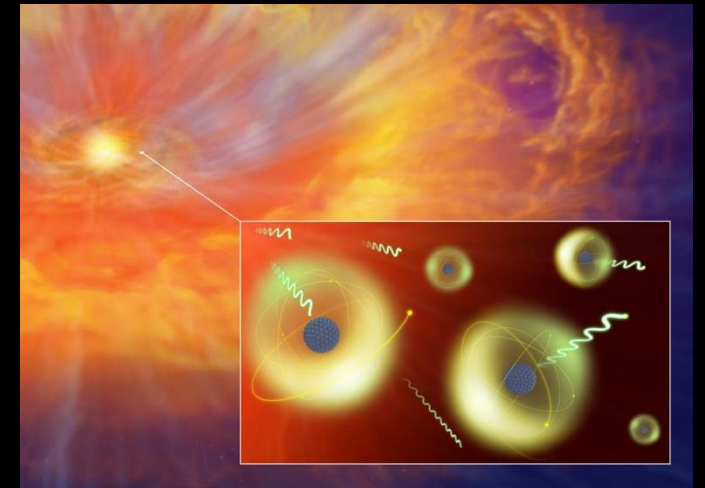
Run	Dates	Duration (months)	Detections
O1	September 18, 2015 to January 12, 2016	3	3
O2	November 30, 2016 to August 25, 2017	9	8
O3	April 1, 2019 to March 26, 2020	11	79
			Candidates
O4	May 24, 2023 to March 19, 2025	23	200

O1 to O3 included about 23 months-worth of observing time between 2015 and 2020. In that time, 90 GW detections were made. In the first 23 months of O4, 200 GW candidates have been logged. O4 is currently scheduled to end in November, 2025.

Why care about double neutron stars?



- Supernova explosions (SNe)
- Gamma ray bursts (GRB) and Kilonovae
- Formation of heavy r-process elements
- Binary stellar evolution



What do we know about Double Neutron Star (DNS) systems so far ?

Detected by radio telescopes and gravitational-wave detectors

Radio Astronomy

~ 26 Galactic DNS in the field with total masses in the range $2.50\text{--}2.89M_{\odot}$

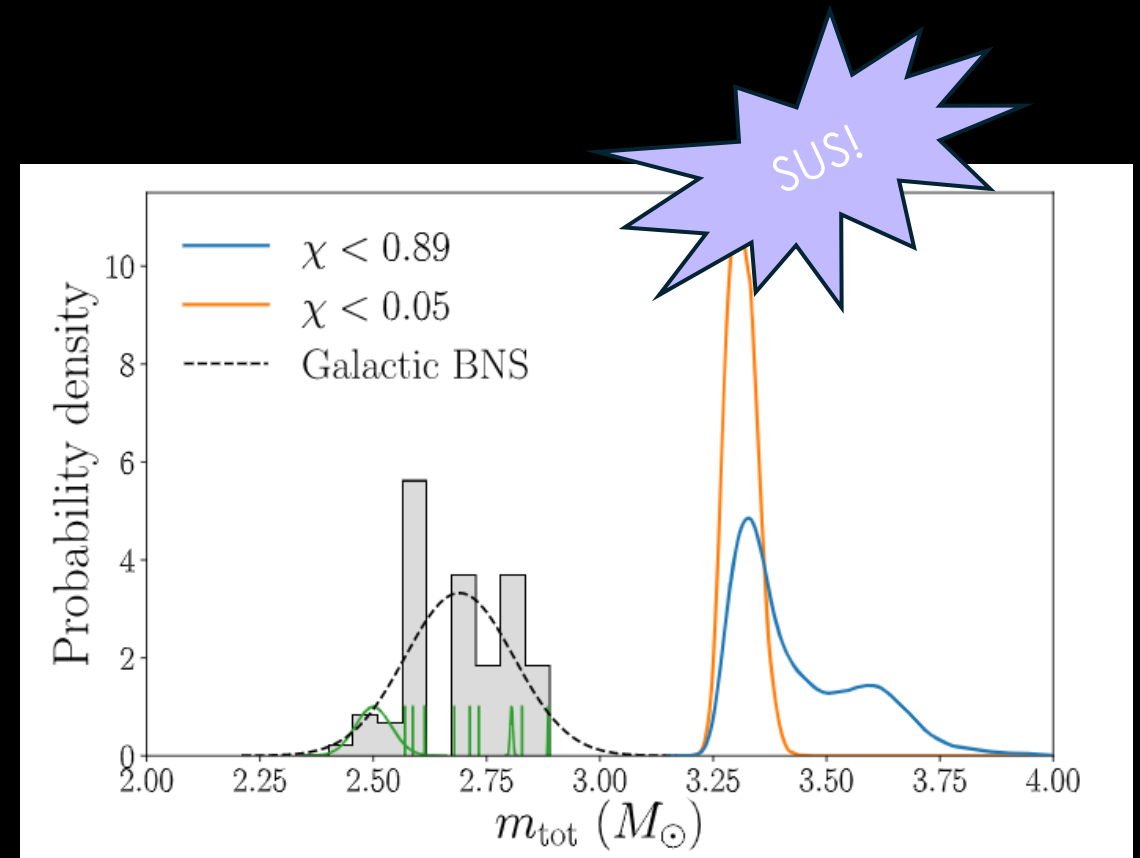
Gravitational Waves

GW170817

Total mass: $2.87^{+0.47}_{-0.09} M_{\odot}$

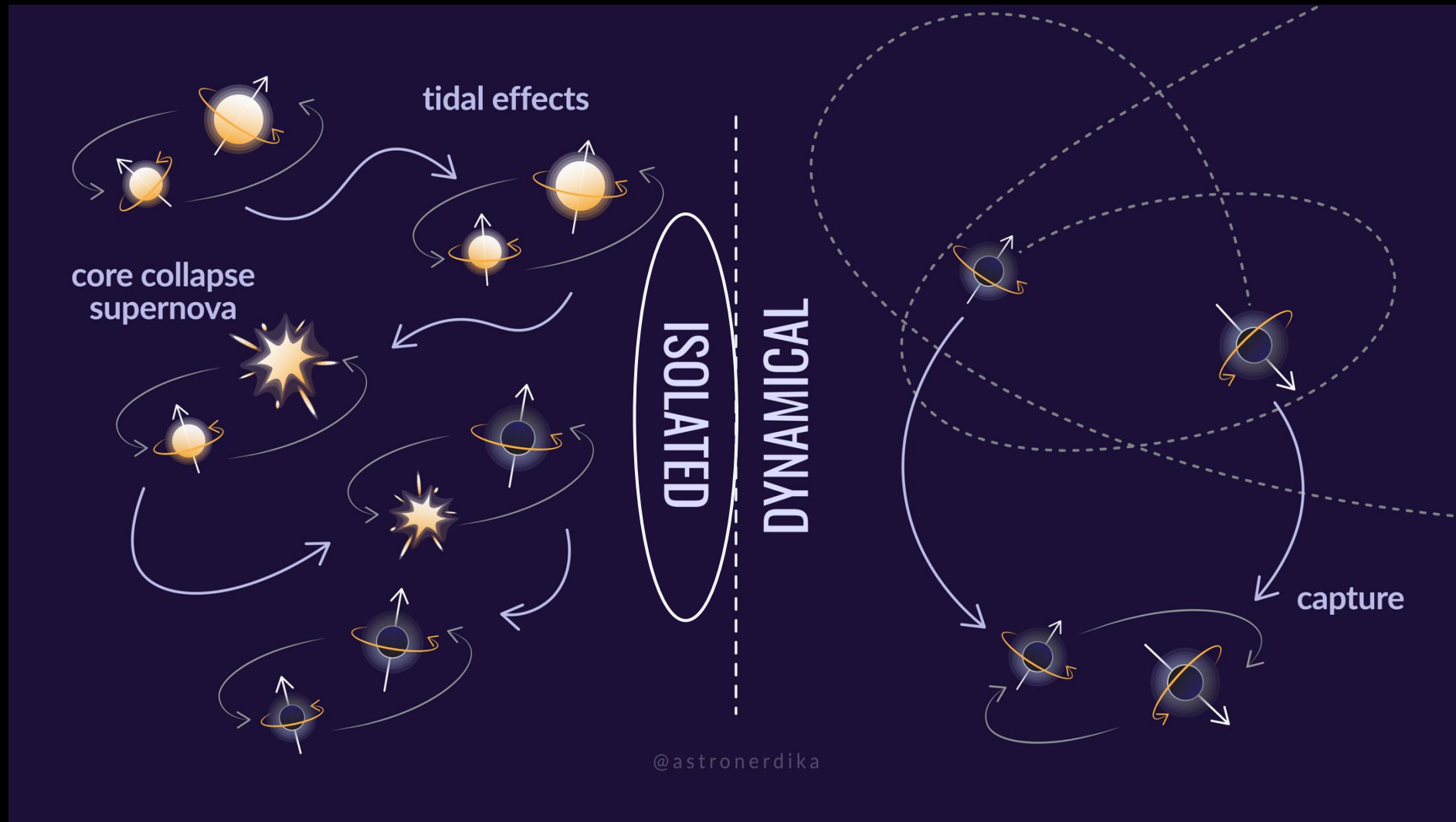
GW190425

Total mass: $3.4^{+0.3}_{-0.1} M_{\odot}$

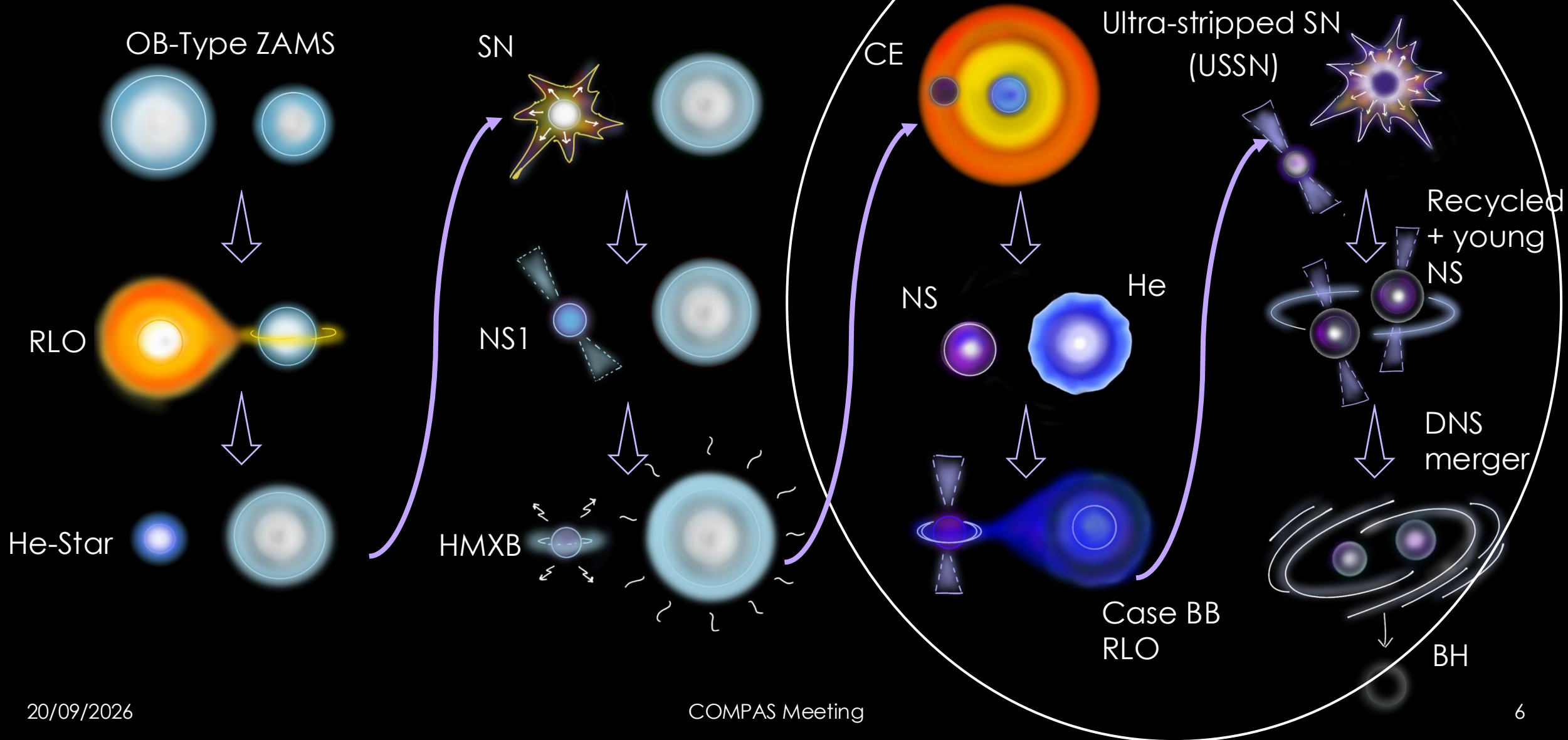


Abbot et al. 2020

Formation Channels of Compact Binary Mergers

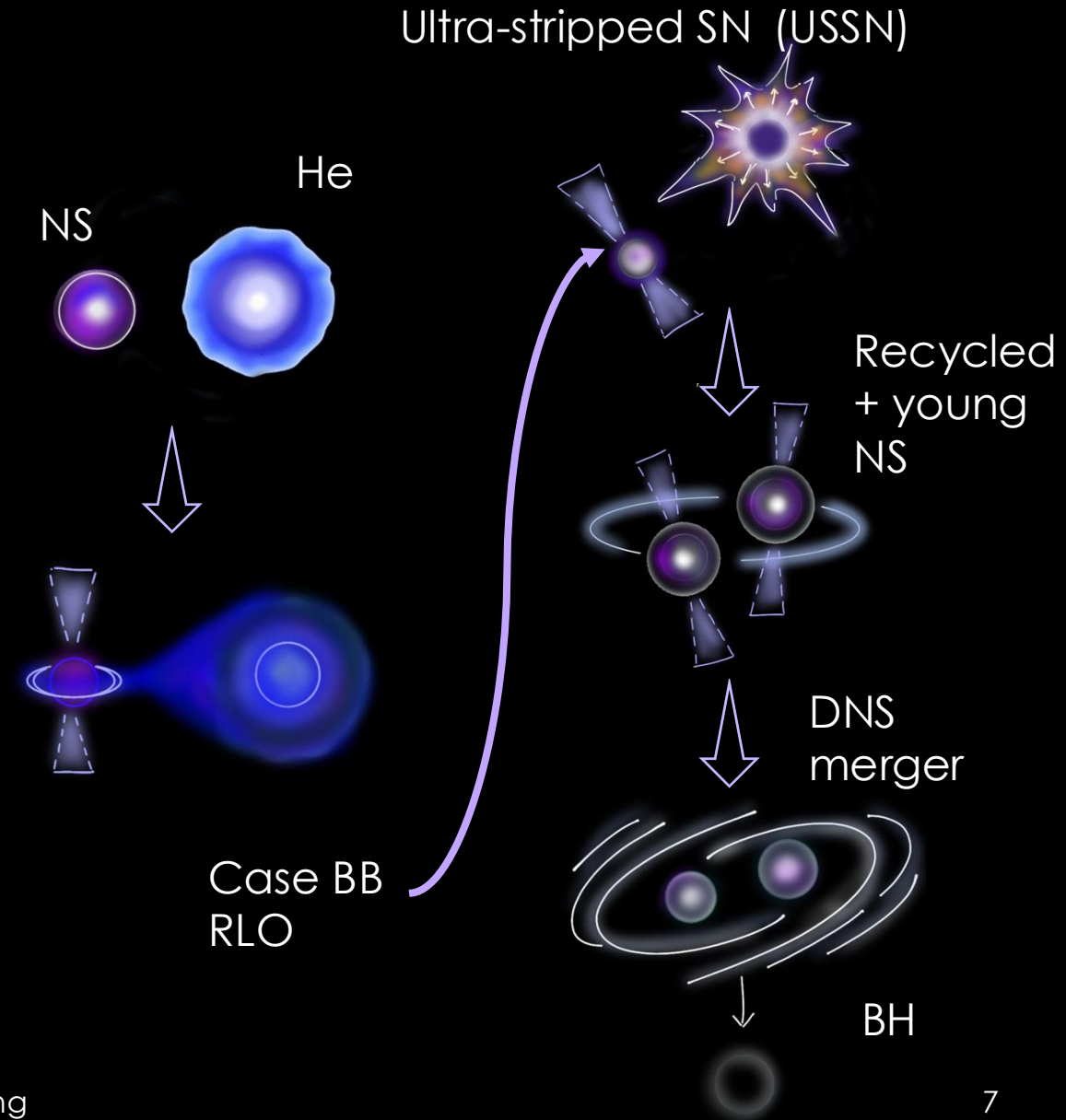


Standard Double Neutron Star Formation Pathway – T. Tauris et al. 2017

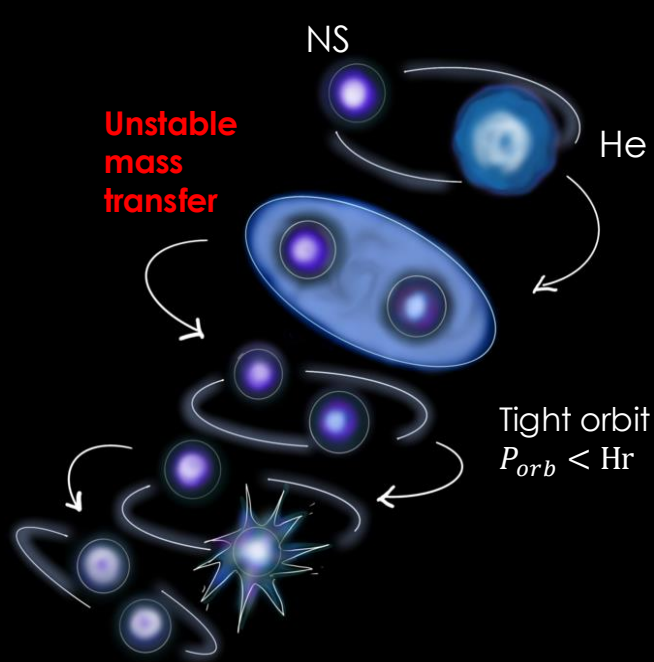


The evolutionary phases that decides the fate of DNSs

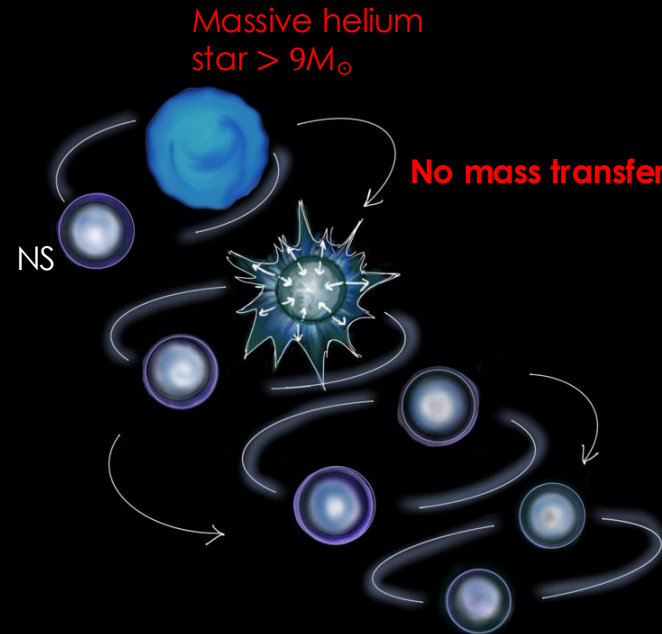
Preferred for modeling DNS formation pathways



Alternate Proposed Formation Pathways for Massive Double Neutron Star



B.P. Abbott et al.
(2020)



Vigna Gómez et al.
(2021)

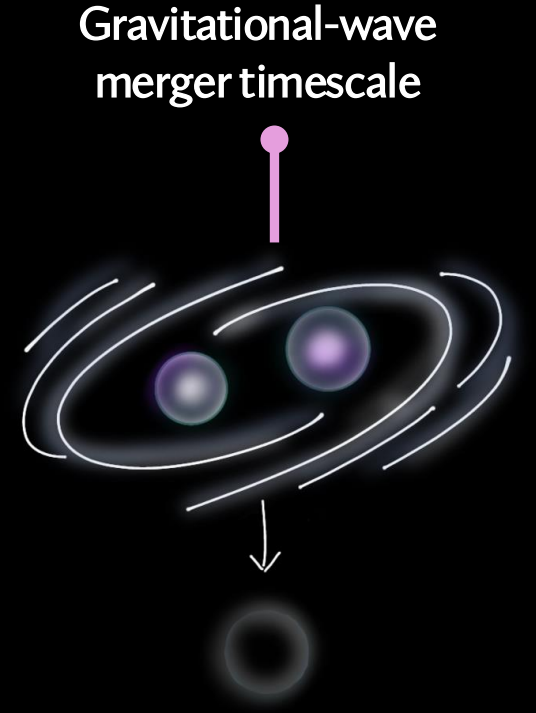
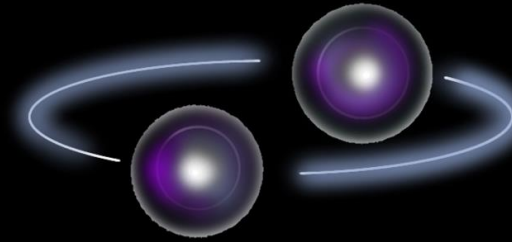
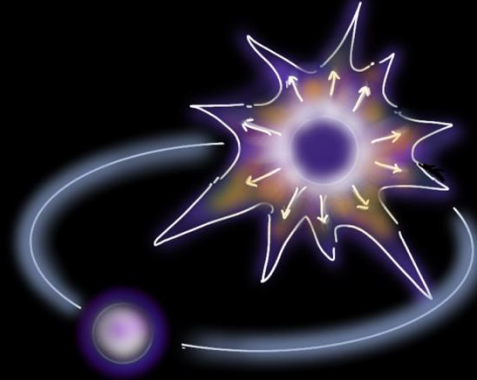
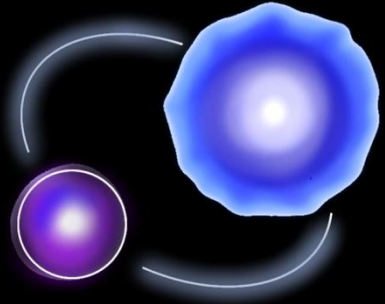
Other uncertainties:

- **Formation at lower metallicities** - Giacobbo & Mapelli (2018)
- **Super-Eddington accretion** - Zhang et al. (2023)
- **Massive first-born NS** - Qin et al (2024)

No self-consistent unbiased formation pathway for heavy double neutron stars

Project I, Part A

Framework Created for the Formation Pathway



ZAMS He (at solar metallicity)-
 $1.4 M_{\odot}$ NS binary evolution

Incorporating supernova and
kick prescription

Post-Supernova effects on
orbital dynamics

Gravitational-wave
merger timescale

MESA

Paxton et al. 2011

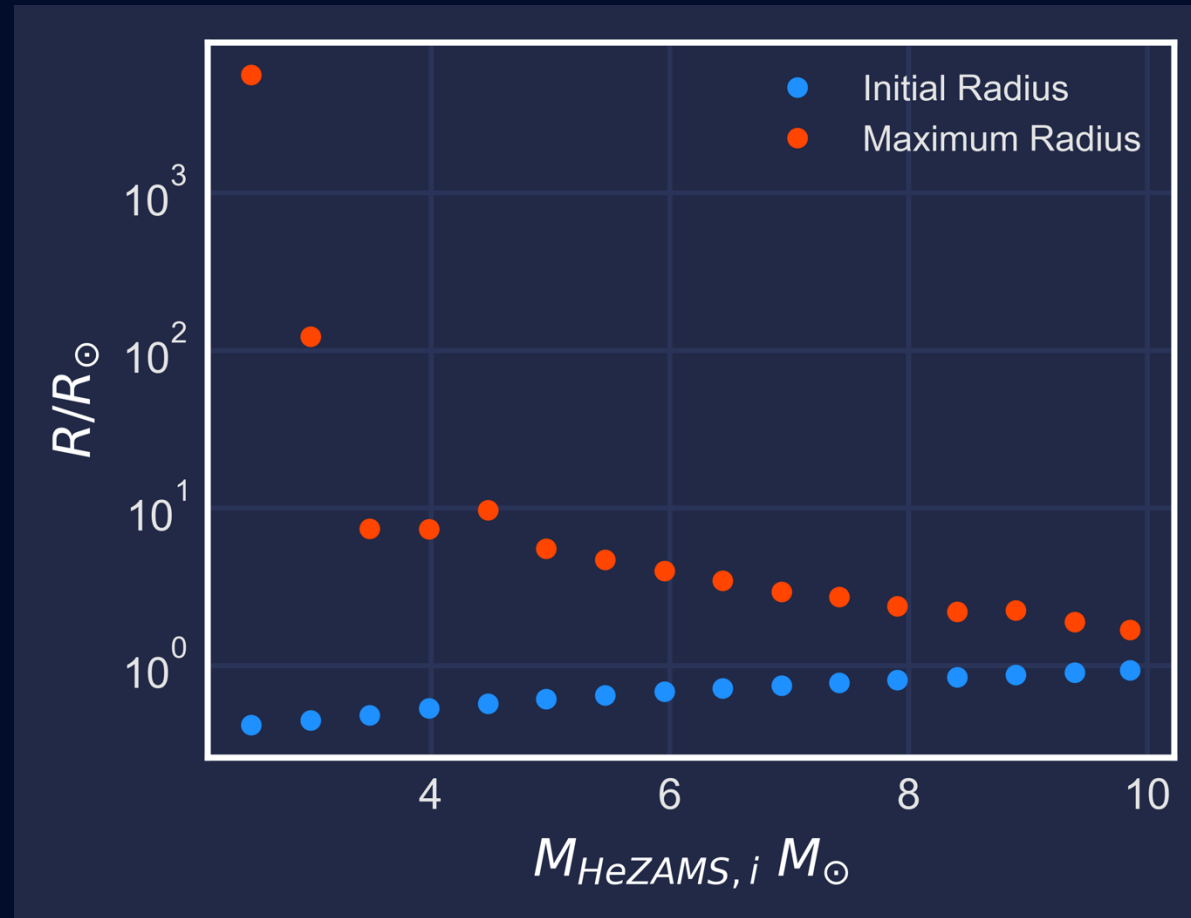
Mandel & Müller 2020,
Kapil et al. 2023

Brandt &
Podsiadlowski 1995

Peters 1964b
Mandel (2021b)

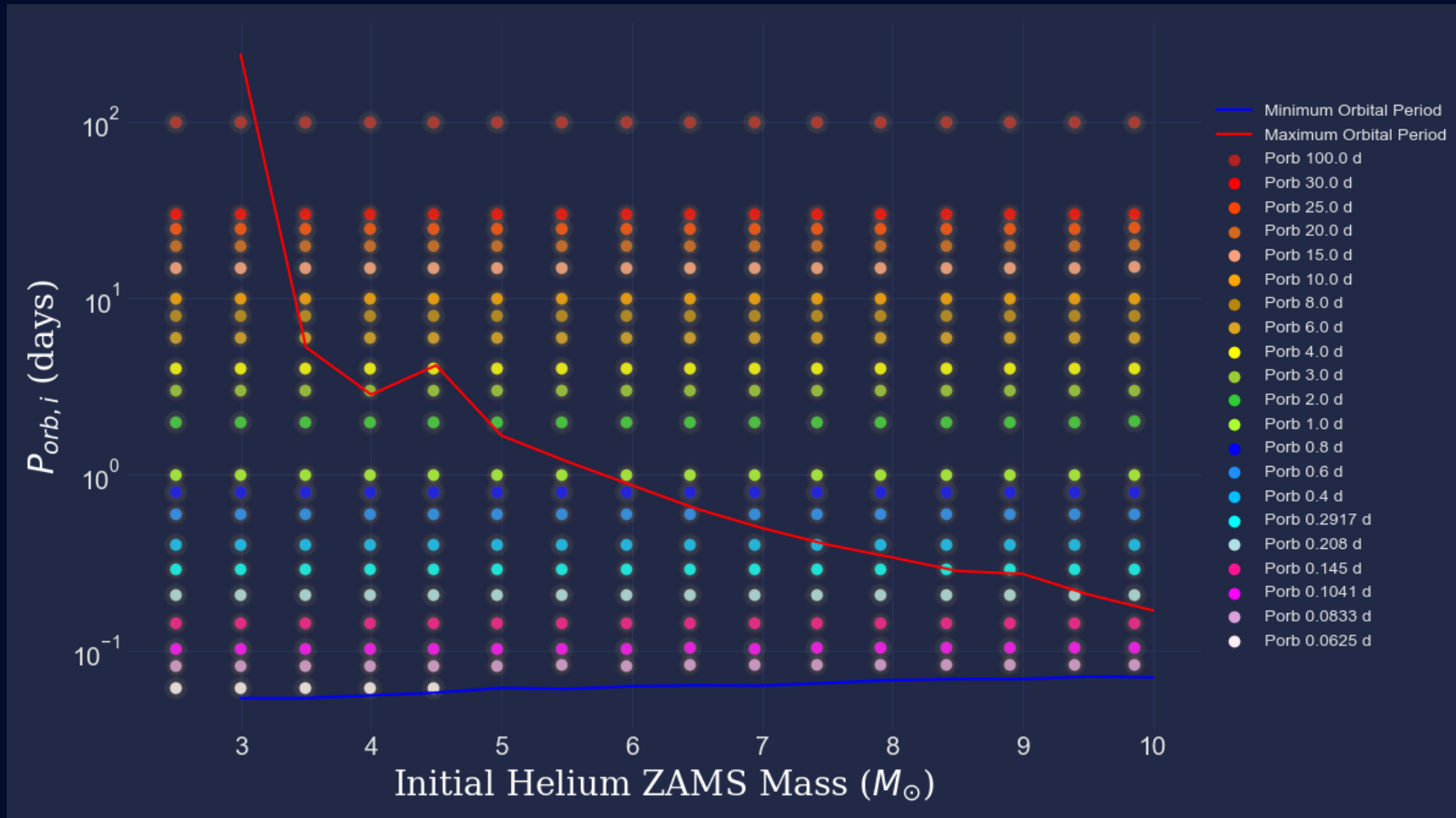
Initial and Maximum Radius as a Function of Initial He-ZAMS Mass at Solar Metallicity

(Recap)



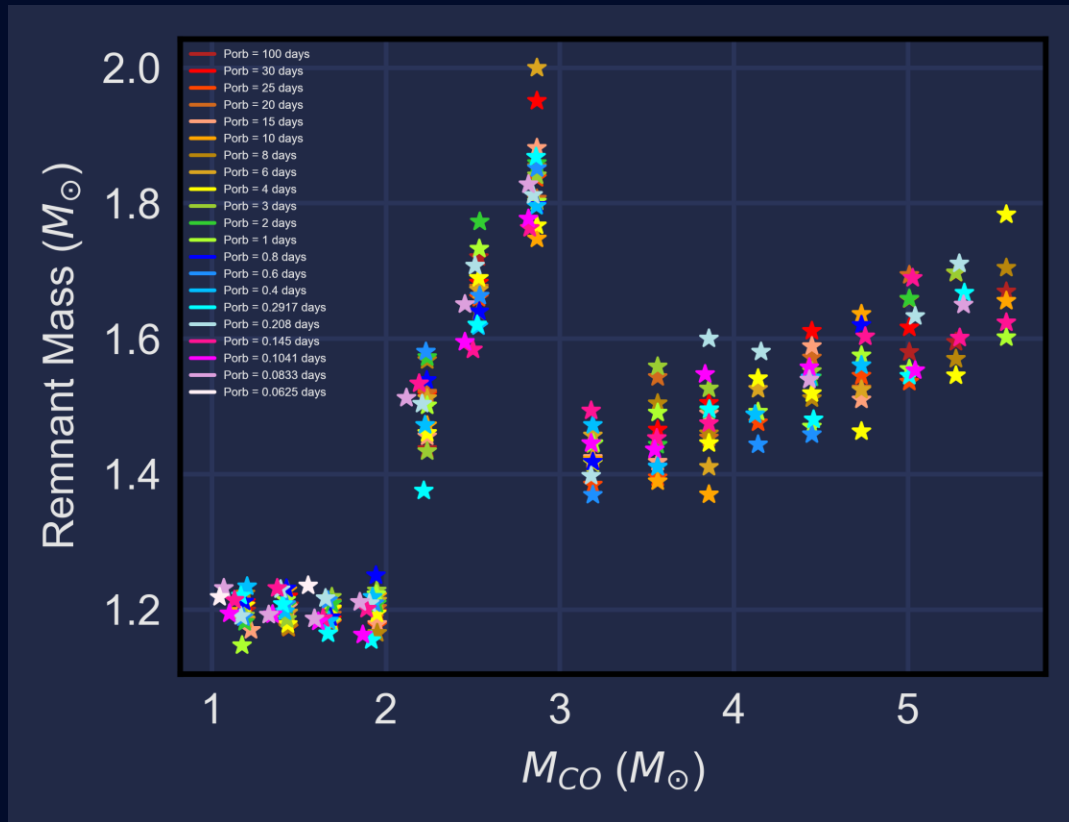
ZAMS :
Zero Age Main Sequence
(aka newly formed stars)

Grid of Models Based On Initial He ZAMS (solar metallicity) Mass + $1.4 M_{\odot}$ NS and Initial Orbital Period

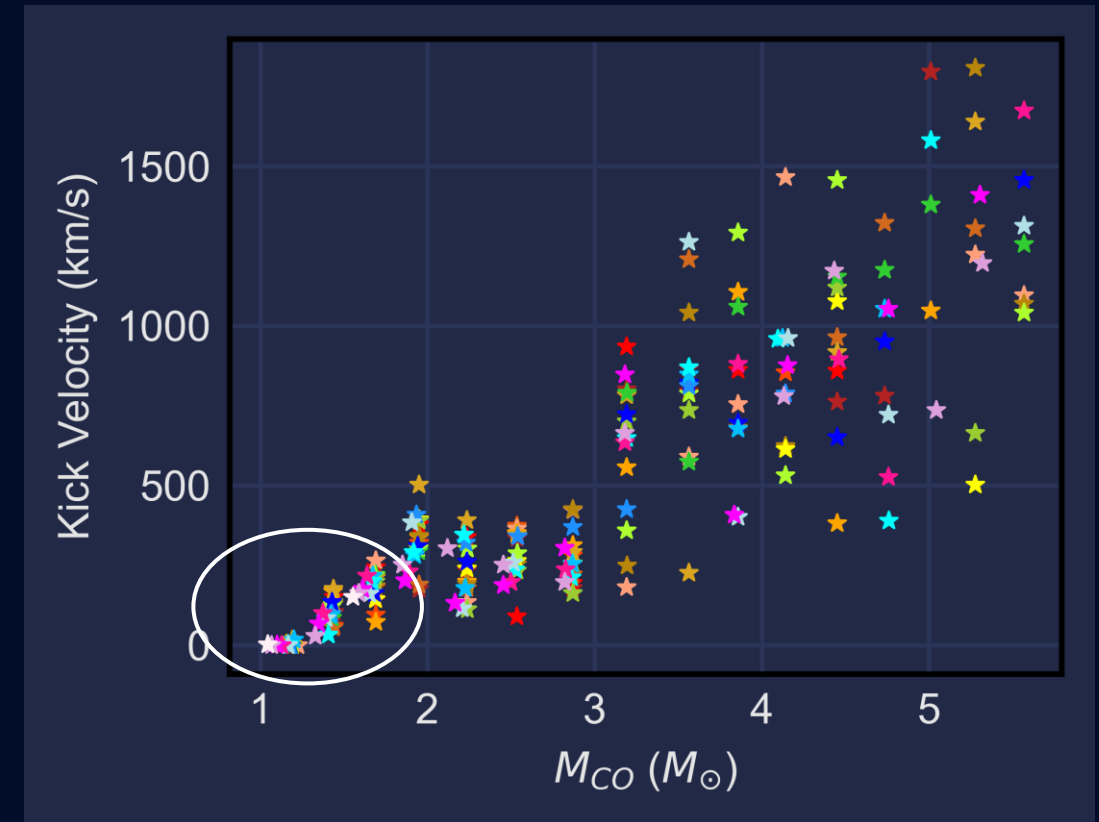




Standard Supernovae and Kick Prescription (Standard Model)



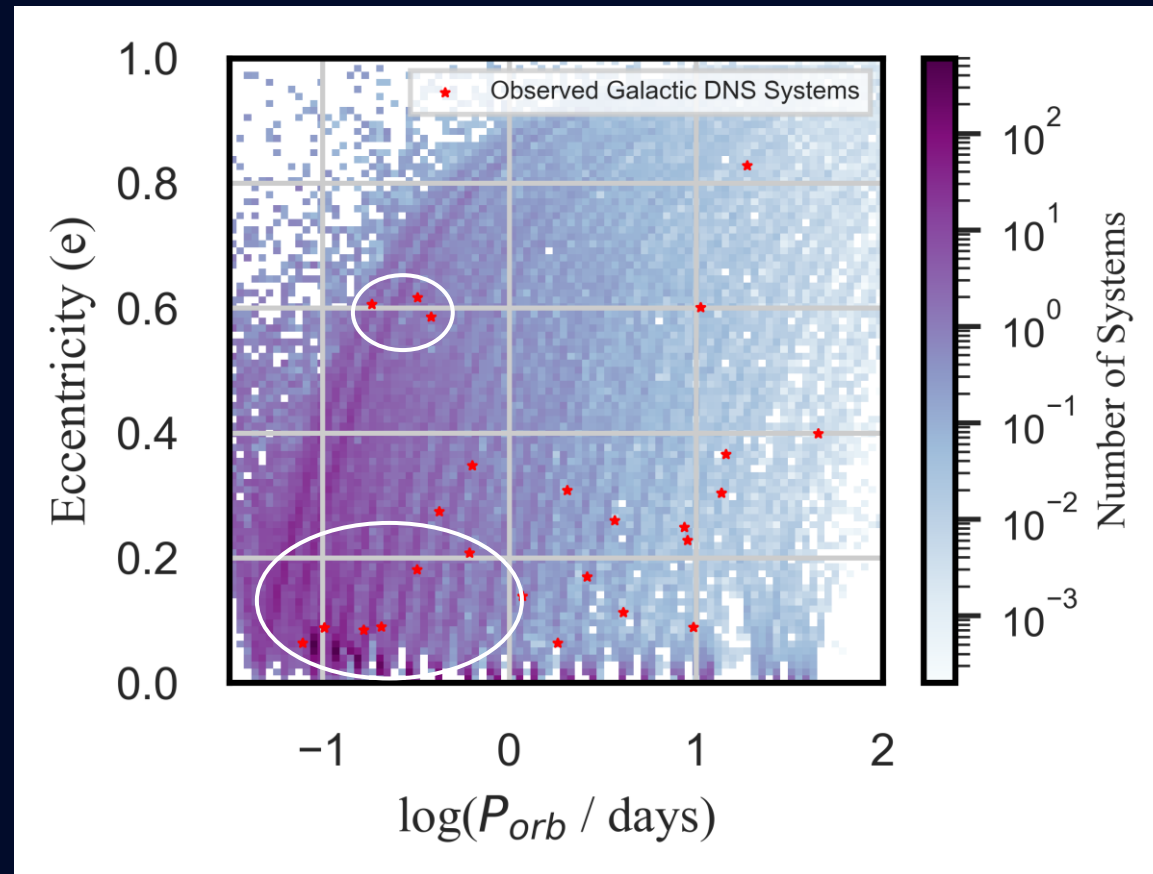
Remnant masses (NS2) as a function of CO core mass



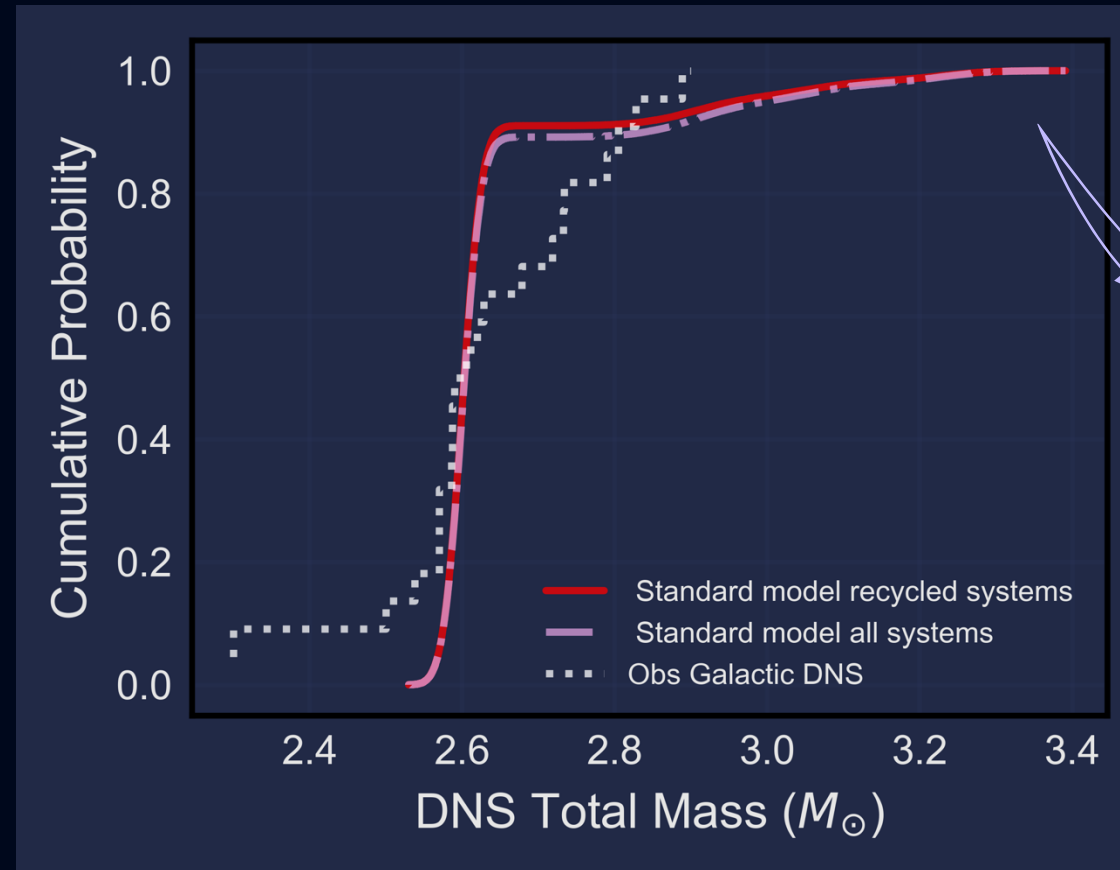
Kick velocity for the corresponding NS2 as a function of CO core mass

Combined P_{orb} -e Distribution of all models

Standard Model



DNS Total Mass Distribution Standard Model



Overproduce high mass systems
than observed

Assumed remnant mass and
CO core mass relation

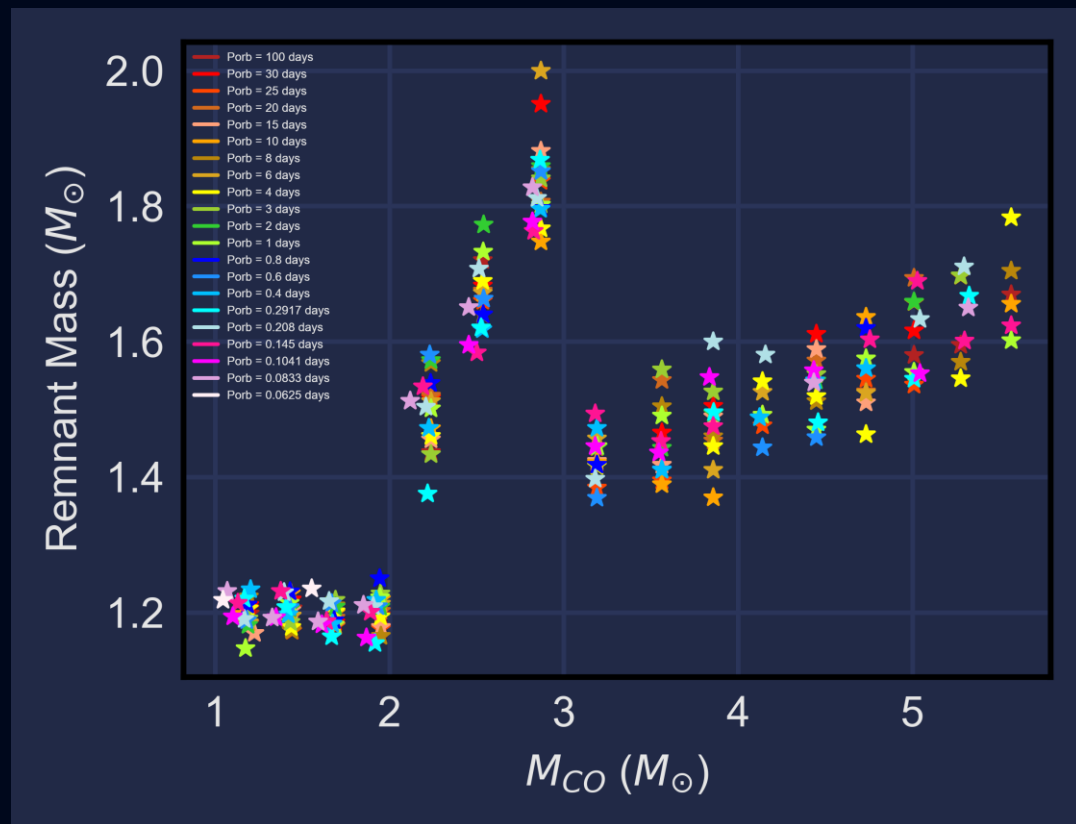
Issues with the Standard Prescription

- Overpredicts orbital eccentricities of the systems
- Overpredicts DNS total mass compared to what we observe at solar metallicity

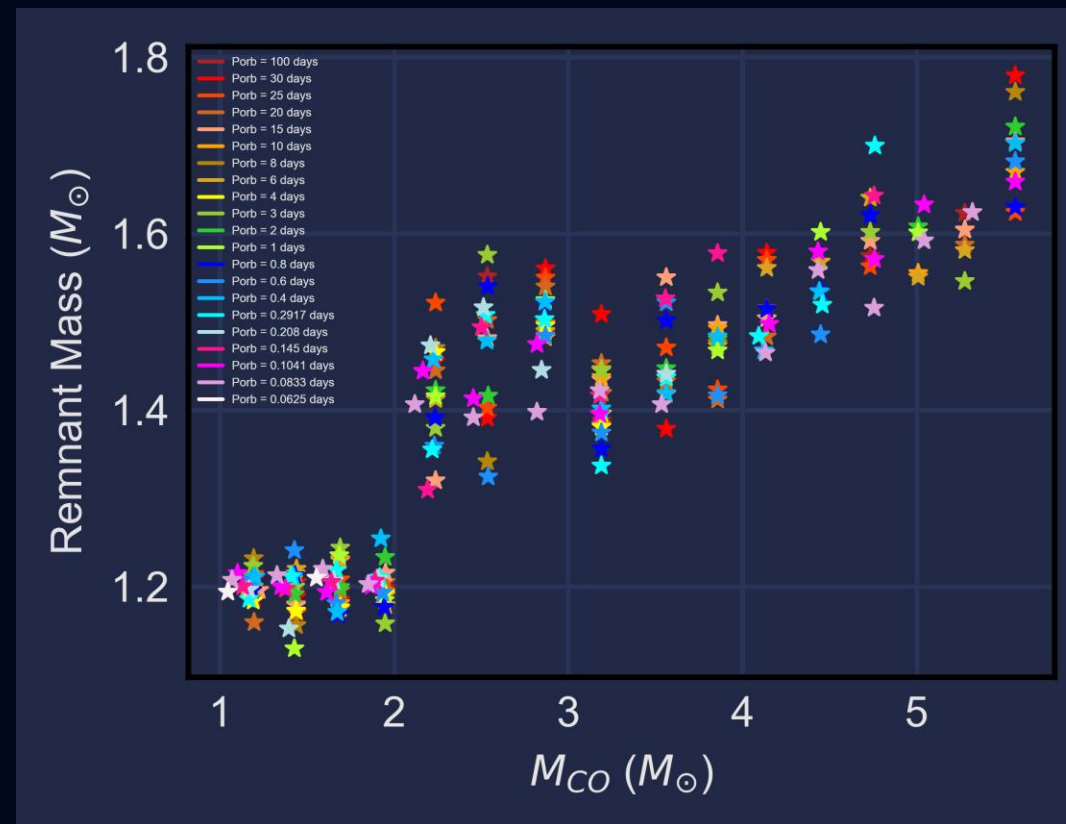
Modifications to the Standard Model

1. Different remnant mass and CO core mass relation
2. Incorporate Ultra-stripped and Electron capture supernova conditions
3. Kick slow merging systems to high velocities

Standard Model (BEFORE)

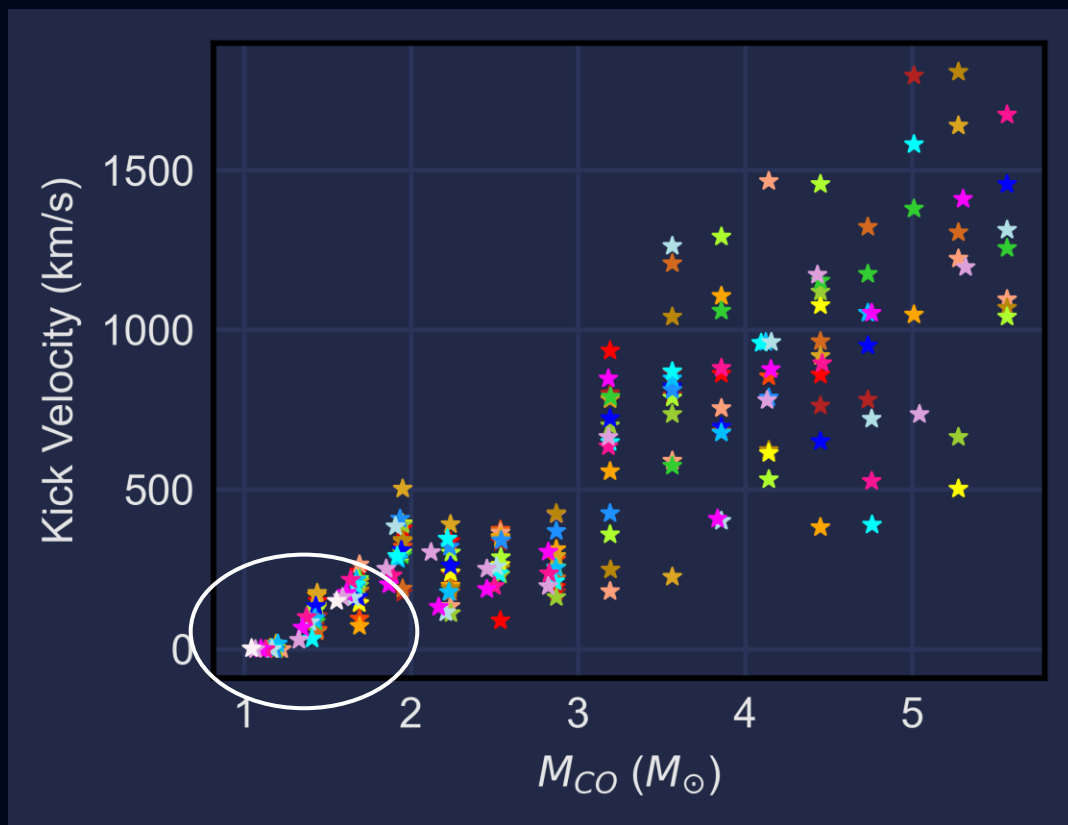


Modified Model

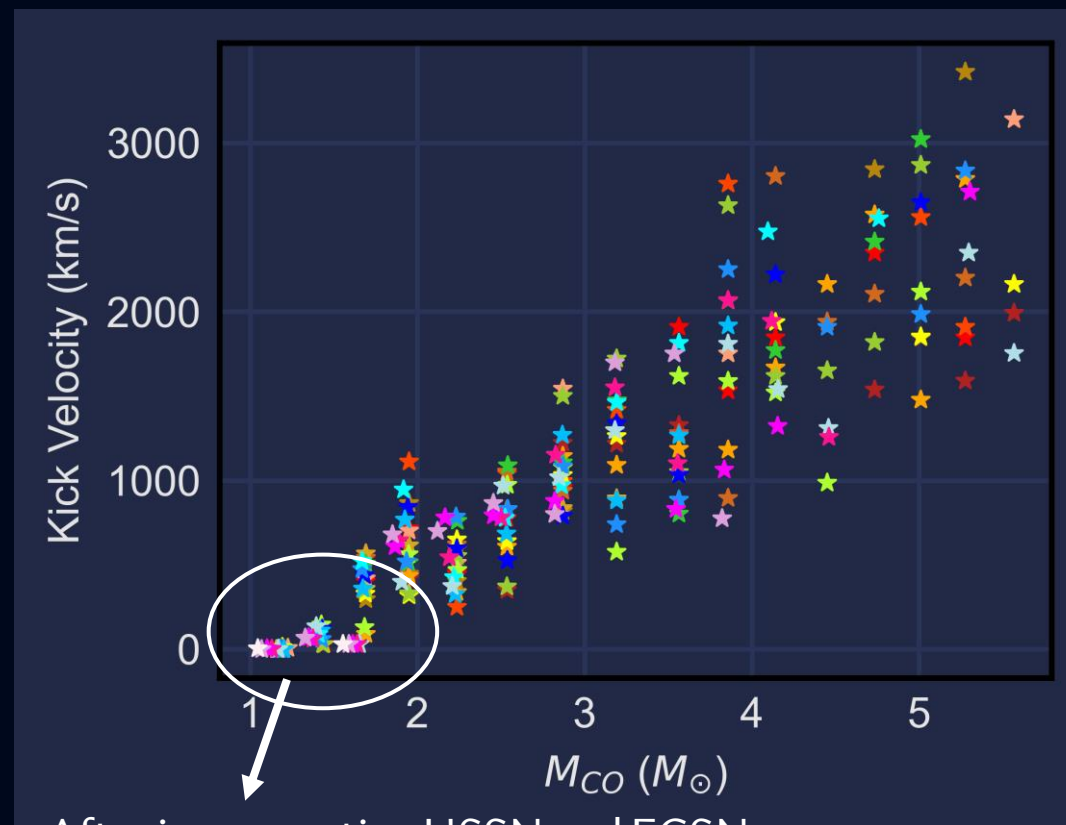


Remnant masses (NS2) as a function of CO core mass

Standard Model
(Before)



Modified Model

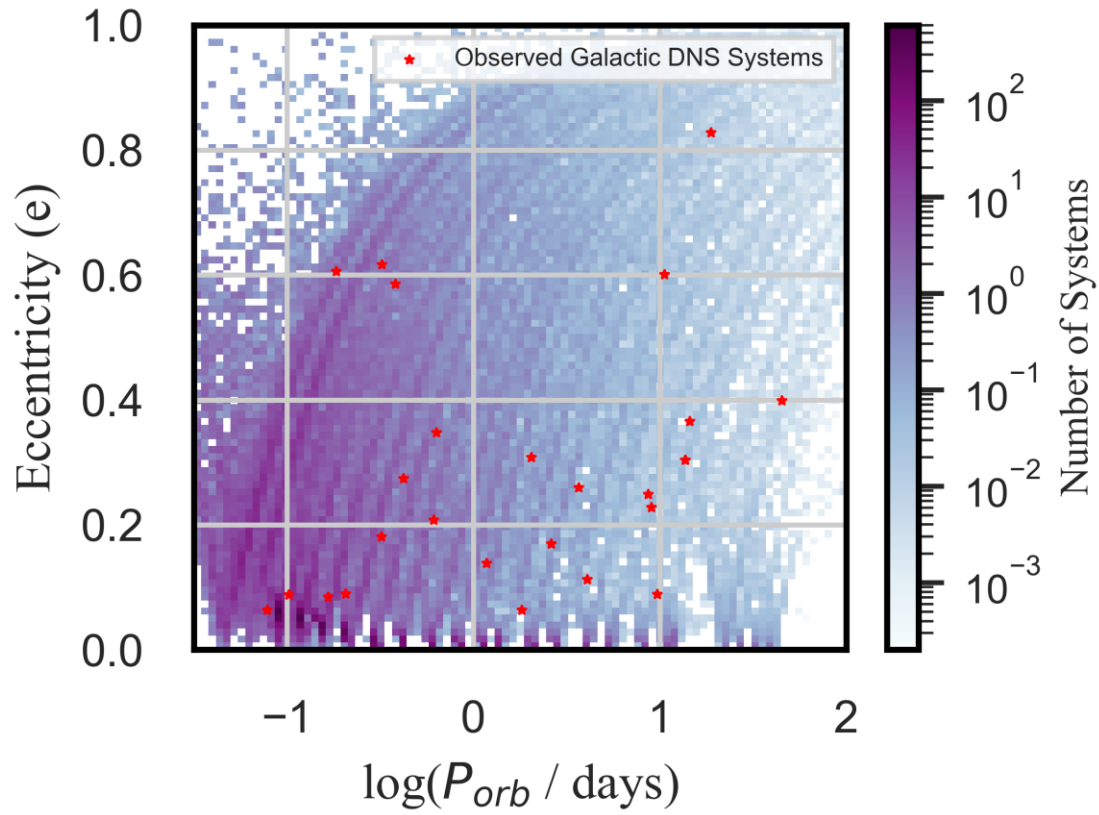


After incorporating USSN and ECSN

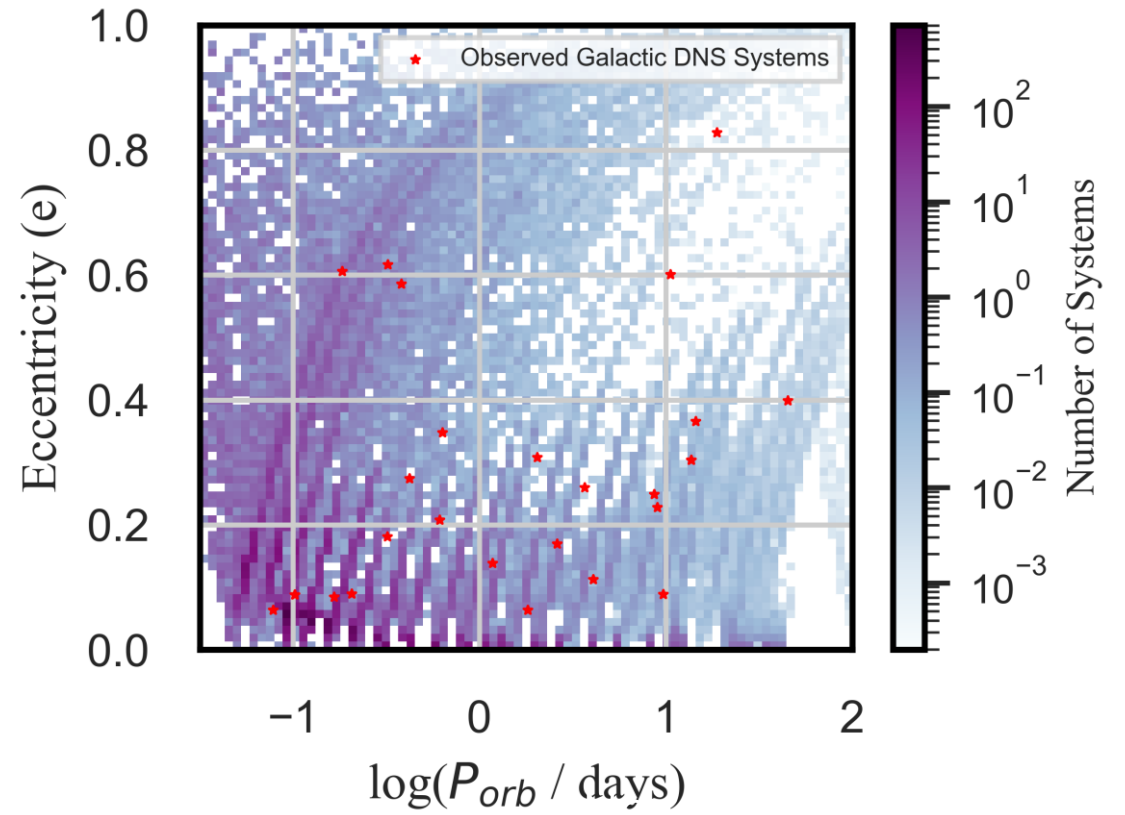
Kick velocity for the corresponding NS2 as a function of CO core mass

Combined Porb-e Distribution of all models

Standard Model
(BEFORE)

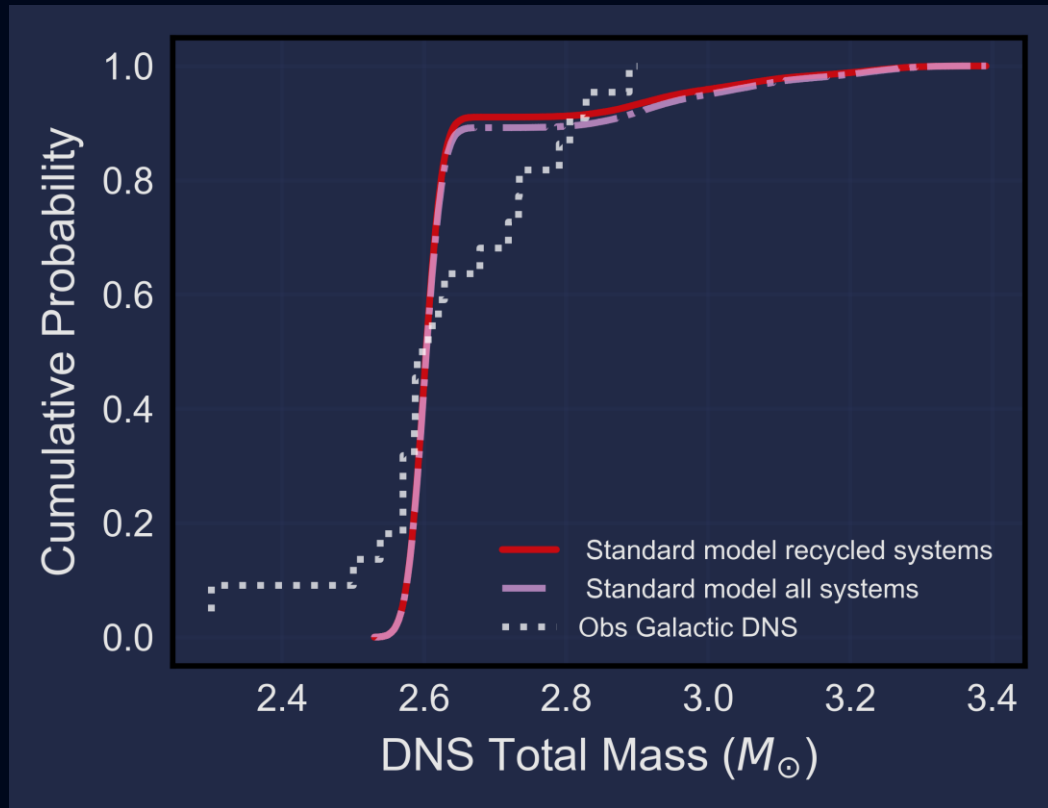


Modified Model

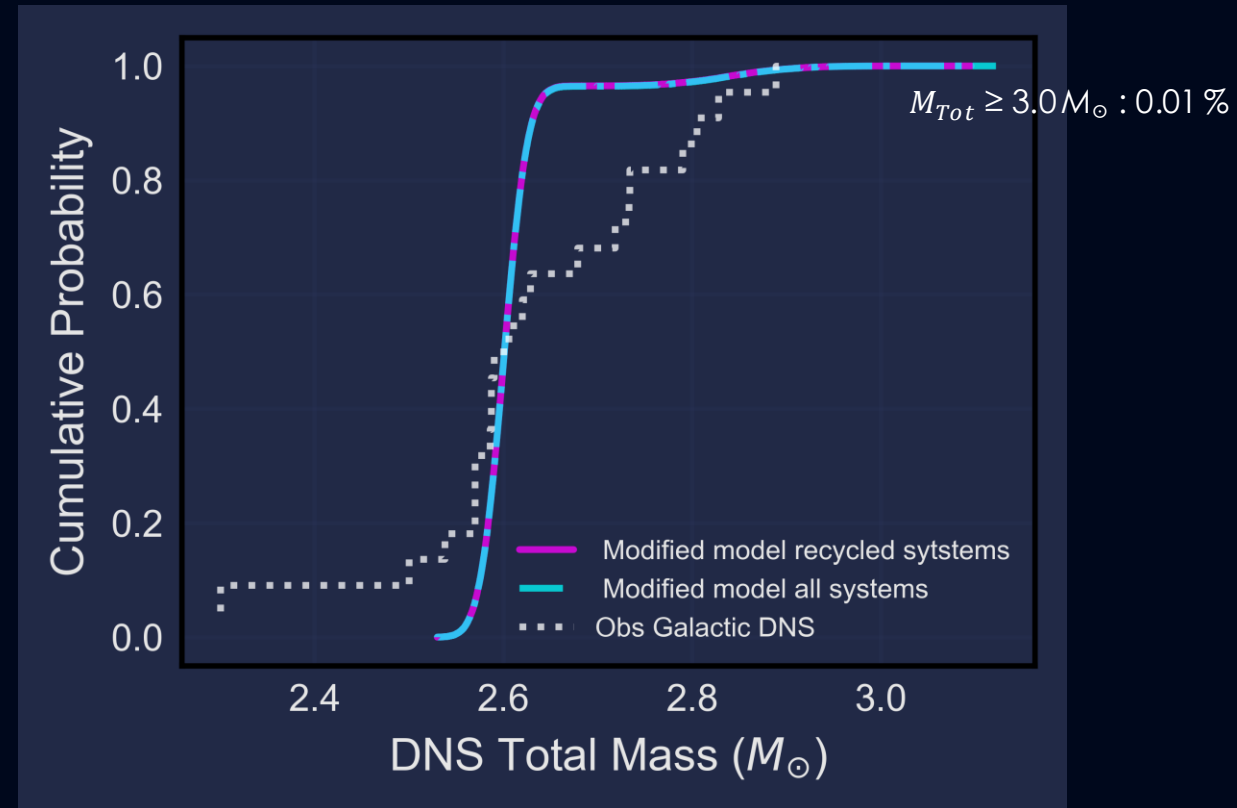


DNS Total Mass Distribution

Standard Model
(BEFORE)



Modified Model

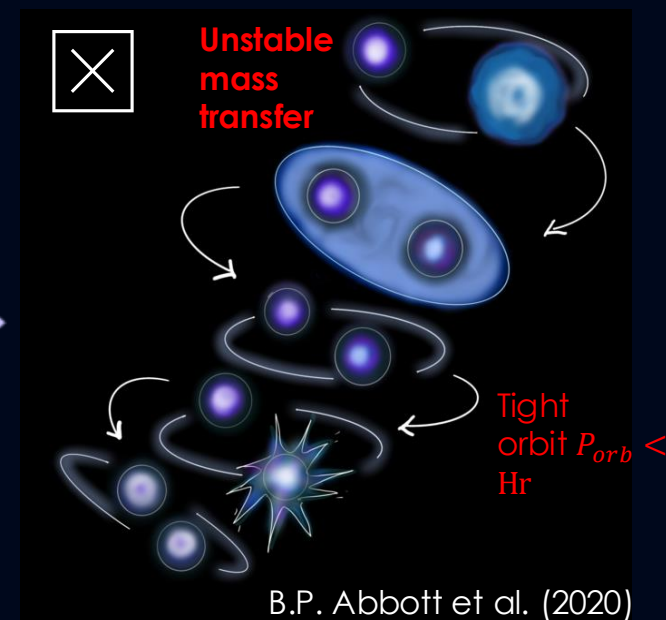
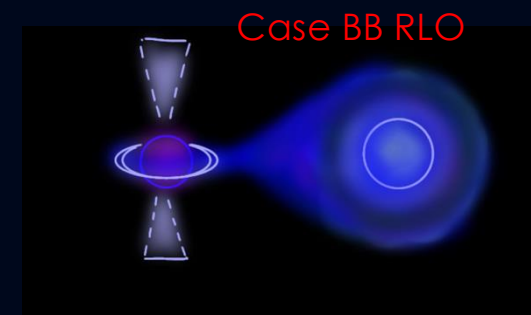


Nair et al. 2025 (In Prep)

GW190425 Progenitors

Modified: (Out of all the systems $\geq 3 M_{\odot}$)

- 82% of systems - formed via Case BB RLO MT
- 18% of systems - formed via massive helium stars without recycling the first-born NS
- Rules out the formation pathway via unstable mass transfer



Fewer DNS systems $\geq 3 M_{\odot}$

Project I, Part B

Explore

- *Low metallicities*
- *Massive first-born NS*
- *Super-Eddington accretion*

Paper I

Investigating the formation pathway of heavy double neutron stars: Part I

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Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

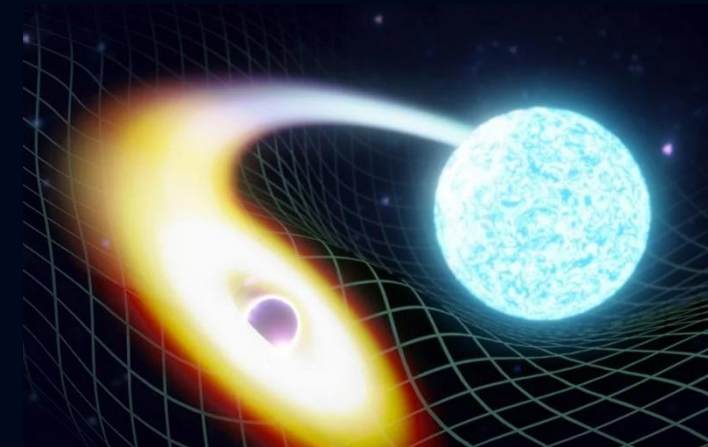
Approximately 30 Galactic double neutron star (DNS) systems have been identified via radio pulsar timing, with total binary masses ranging from 2.50 to 2.89 $M_{\odot}$. Gravitational-wave (GW) observations have yielded two detections of binary neutron star (BNS) mergers so far: GW170817, with a total binary mass of 2.74 $M_{\odot}$ and GW190425, with a combined mass of 3.4 $M_{\odot}$. The total mass of GW190425 is significantly higher than any known Galactic DNS system, suggesting a non-standard formation pathway. To investigate the formation of heavy double neutron star binaries, we explore the late stages of evolution of helium stars at solar metallicity with masses between 2.5 and 9.8 $M_{\odot}$ in binary systems with a 1.4 $M_{\odot}$ neutron star companion, modeled using the detailed 1D stellar evolution code — Modules for Experiments in Stellar Astrophysics (MESA). We test proposed alternative formation channels to determine which pathway(s) are more likely to produce heavy systems like GW190425. We calibrate our models to ensure they can reproduce the observed Galactic DNS population, including their mass distribution and orbital properties. Our binary models, evolved using MESA with a modified kick prescription, can roughly explain the observed DNS total mass distribution. We conclude that the population synthesis of our models rules out the formation of heavy DNS systems through the second unstable mass transfer phase.

Key words: gravitational waves – stars: massive – stars: evolution – binary evolution

(To be submitted)

Project II

- Studying the formation pathway of any double neutron star (DNS) system detected during the ongoing LVK O4 run
- Analyzing the formation pathways and rates of simulated neutron star–black hole (NSBH) systems, and comparing them with all NSBH candidates detected via gravitational waves.



NSBH binary

Conclusions



Models calibrated to match the Galactic DNS population



Roughly reproduces the observed Galactic DNS distribution



Used models to explore **GW190425 progenitors**



Massive systems (like GW190425) formed at **solar metallicity** are **rare**



Unstable Case BB MT pathway does not yield GW190425-like binaries



Model limitations: Can't explain high GW190425 rates



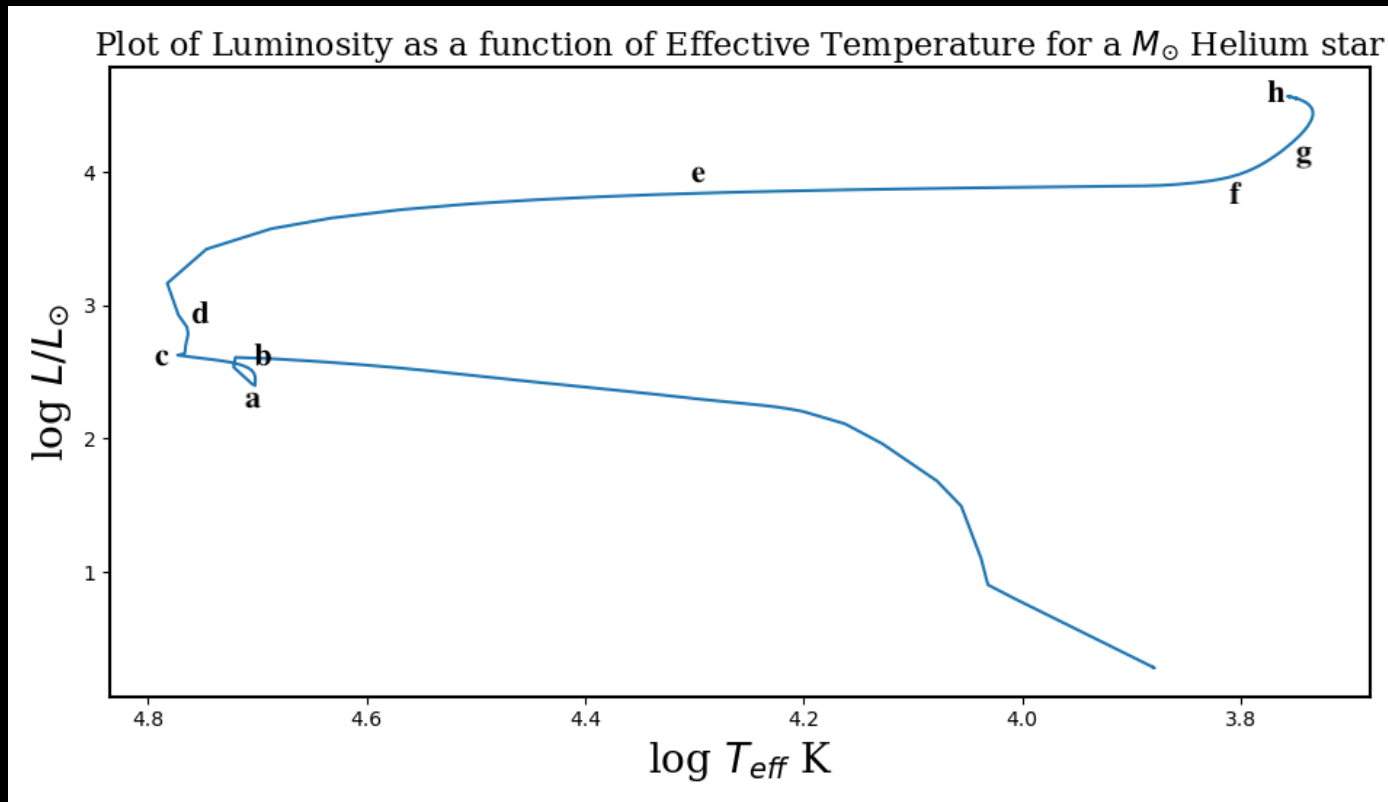
Next steps: Explore low-Z models, massive NS primaries & super-Eddington accretion



Hoping O4 run detects at least one DNS

EXTRA SLIDES

1 $M_{\odot}$ Helium Star Evolution



a : zero age main sequence

a - b : Main sequence phase

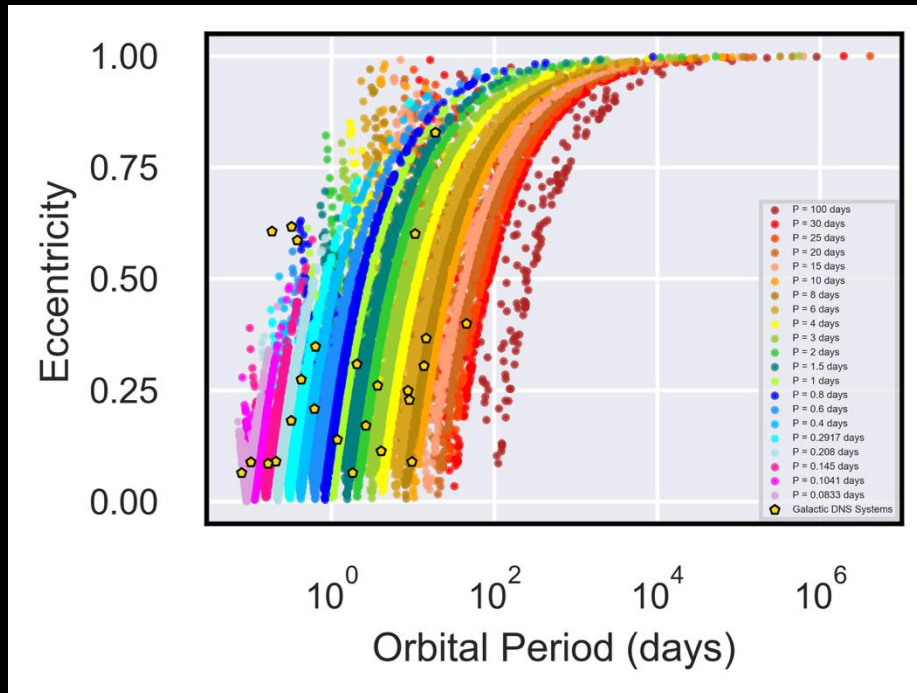
b - c : exhaustion of central helium,

c - d : first cooling phase,

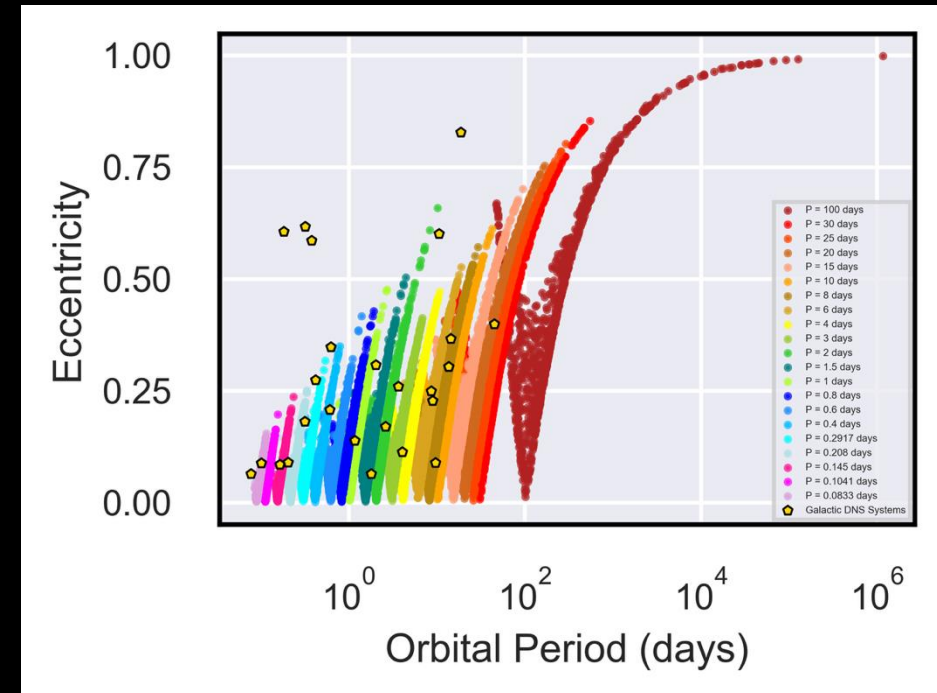
d - e : slow phase of shell helium burning

e - h : fast phase of helium shell burning

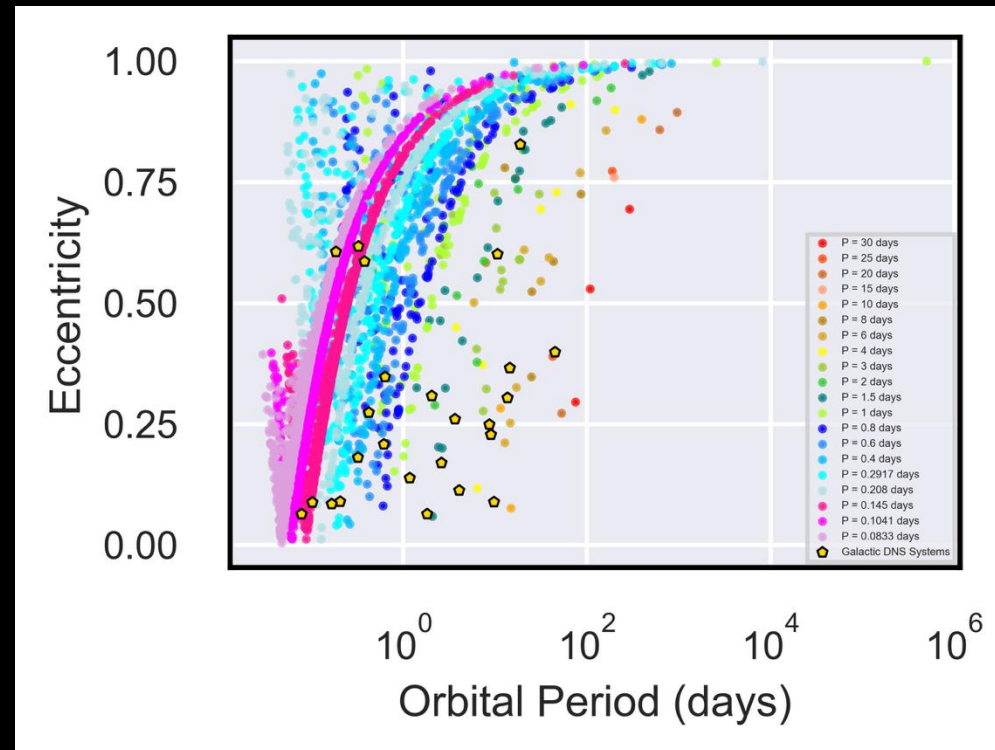
Standard model 3Msun



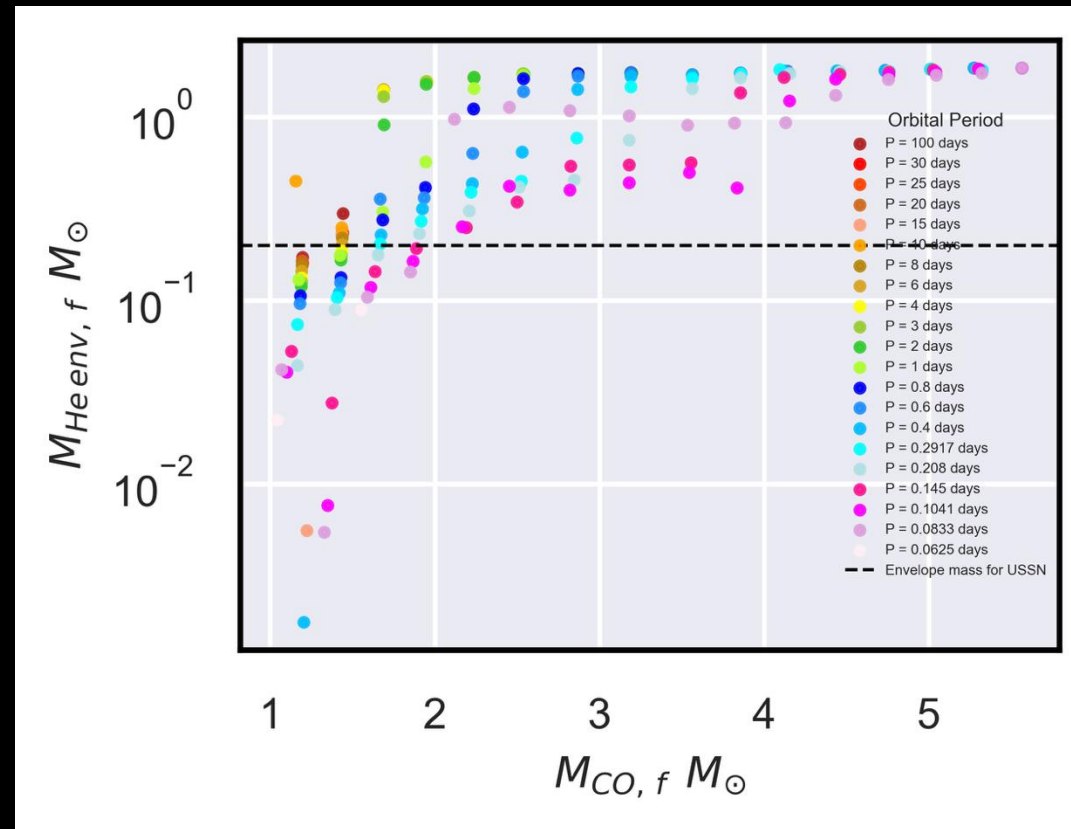
Modified 3Msun



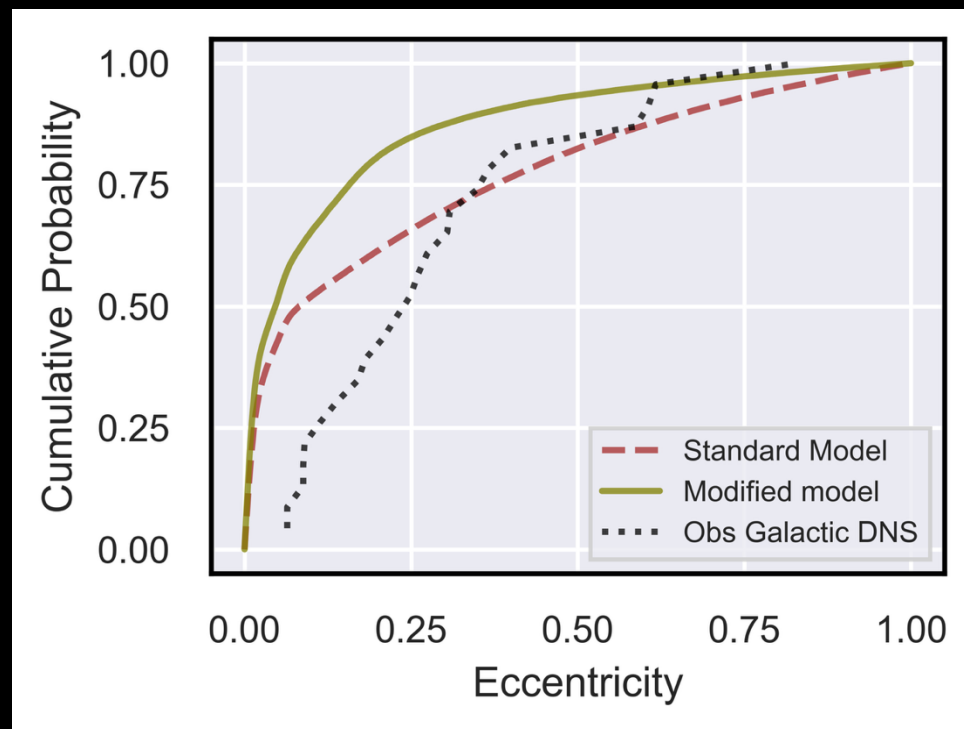
Modified 4Msun



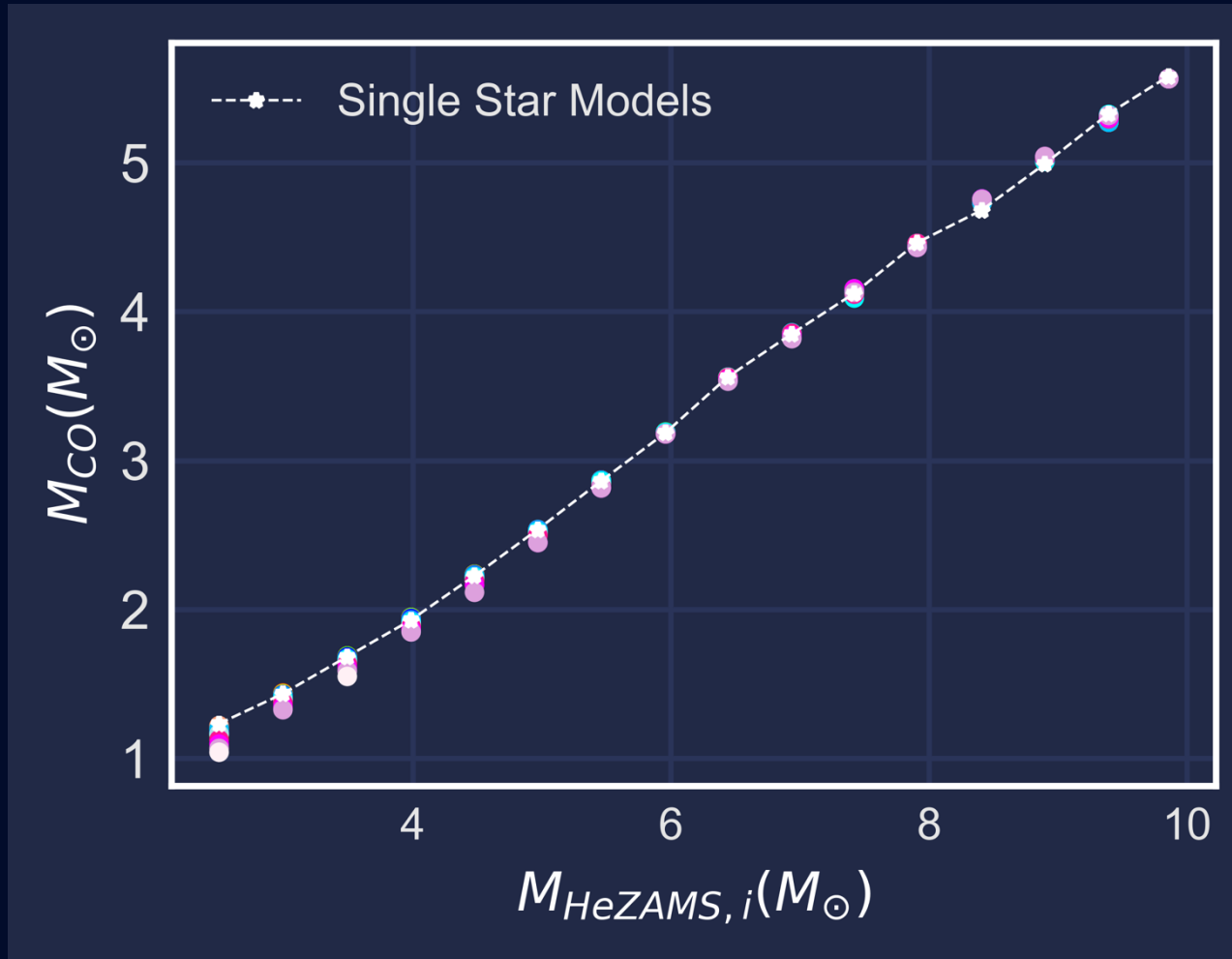
Final helium envelope pre-SN



Eccentricity



Carbon-oxygen (CO) core masses as function of initial He-ZAMS mass



He-NS

mass

separation

Mass transfer, accretion

metallicities

First being the massive one

CO core/ metal relation and explosibility

SN Explosion

Kick velocities

Types/prescription/assumptions

Why high kick for fast merging systems :

We consider the hypothesis that the high-mass gravitational-wave event GW190425 is part of a subpopulation formed through unstable case BB mass transfer, which quickly merge in $\sim 10\text{--}100$ Myr. Shanika G n Simon

Romero-Shaw et al. (2020) argue that GW190425 is best understood as a fast-merging binary, which underwent unstable case BB mass transfer. They speculate that GW190425 could have formed from a progenitor including a $4\text{--}5M_{\odot}$ He star with a $\sim 3M_{\odot}$ CO core capable of producing a $\sim 2M_{\odot}$ neutron star (Müller et al. 2016; Ertl et al. 2020). According to this hypothesis, massive DNS like GW190425 are not observed in the radio because they merge before they are likely to be seen.

This fast-merging hypothesis seems to rely on the premise that the most massive slow-merging DNS are disrupted by large supernova kicks. In the appendix, we present the results of a population synthesis study to show that large kicks $\sim 2000\text{ km s}^{-1}$ are required for this theory to have explanatory power.

Papers and reason for linear co core and remnant mass relationship – see whats there argument for the same – gilgis et al. 2025

Post-sn eccentricities and orbital period

Kick and orbit alteration